2025-10-17 - Valentini (6)

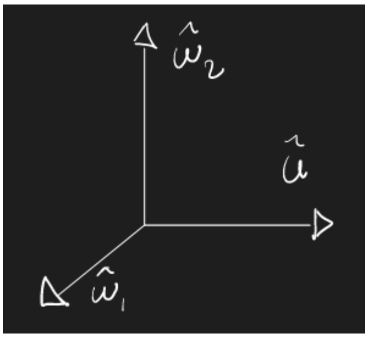
Previous lecture: 2025-10-16 - Valentini (5)
Next lecture: 2025-10-23 - Servidio (7)

$$k^2\tilde{\phi}_{\bar{k}}(p) = \frac{4\pi \sum_\alpha q_\alpha \int_{-\infty}^{\infty} \frac{f_{\alpha\bar{k}}(v, t=0)}{p + i\bar{k}\cdot\bar{v}}\, d^3v}{1 - \frac{4\pi}{k^2} i \sum_\alpha \frac{q_\alpha^2}{m_\alpha} \int_{-\infty}^{\infty} \frac{\bar{k}\cdot\nabla_{\bar{v}} f_{\alpha 0}}{p + i\bar{k}\cdot\bar{v}}\, d^3v}$$

we want to antitransform this

$$\phi_k(t) = \frac{1}{2\pi i}\int_{p_0 - i\infty}^{p_0 + i\infty} \exp(pt)\tilde{\phi}_k(p)\, dp$$

We rotate the reference frame such that one of the new axis is parallel to $\bar{k}$
The new reference frame will be $(\hat{u}, \hat{w}_1, \hat{w}_2)$ with $\hat{u} \parallel \bar{k}$



$$u = \frac{\bar{k}\cdot\bar{v}}{k} \implies \bar{k}\cdot\bar{v} = ku \implies \bar{k}\cdot\nabla_{\bar{v}} = k\frac{\partial}{\partial u}$$

so

$$k^2\tilde{\phi}_{\bar{k}}(p) = \frac{4\pi \sum_\alpha q_\alpha \int_{-\infty}^{\infty} \frac{f_{\alpha\bar{k}}(u, w_1, w_2)}{p + iku}\, du\, dw_1\, dw_2}{1 - \frac{4\pi}{k^{\cancel{2}}} i \sum_\alpha \frac{q_\alpha^2}{m_\alpha} \int_{-\infty}^{\infty} \cancel{k} \frac{\partial f_{\alpha 0}(u, w_1, w_2)/\partial u}{p + iku}\, du\, dw_1\, dw_2}$$

The poles are only along u since

$$p = -iku \implies u = \frac{ip}{k}$$

We define (at $t = 0$)

$$F_{\alpha 0}(u) = \int_{-\infty}^{\infty} f_{\alpha 0}(u, w_1, w_2)\, dw_1\, dw_2$$

$$F_{\alpha\bar{k}}(u) = \int_{-\infty}^{\infty} f_{\alpha\bar{k}}(u, w_1, w_2)\, dw_1\, dw_2$$

(we assume f as having all the properties for the integral to be stable).
We substitute into ϕ and

$$k^2\tilde{\phi}_{\bar{k}}(p) = \frac{4\pi \sum_\alpha q_\alpha \int_{-\infty}^{\infty} \frac{F_{\alpha\bar{k}}(u)}{p+iku}\, du}{1 - \frac{4\pi}{k} i \sum_\alpha \frac{q_\alpha^2}{m_\alpha} \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{p+iku}\, du} =$$

$$= \frac{(-i)\frac{4\pi}{k} \sum_\alpha q_\alpha \int_{-\infty}^{\infty} \frac{F_{\alpha\bar{k}}(u)}{u-ip/k}\, du}{1 - \frac{4\pi}{k^2} \cancel{i} \sum_\alpha \frac{q_\alpha^2}{m_\alpha} \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u-ip/k}\, du}$$

where in the last passage we collected ik in both the integrals

The denominator is the **Electrostatic Dielectric Function**

$$D(k, ip) \equiv 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u-ip/k}\, du$$

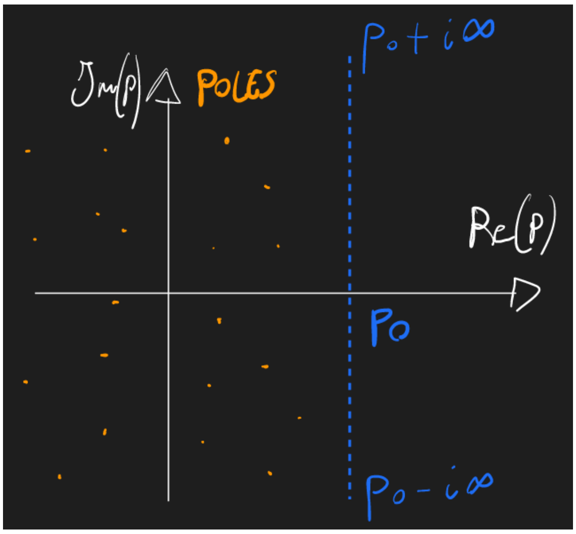
and

$$\phi_k(t) = \frac{1}{2\pi i} \int_{p_0-i\infty}^{p_0+i\infty} \frac{(-i)\frac{4\pi}{k}\sum_\alpha q_\alpha \int_{-\infty}^{\infty} \frac{F_{\alpha\bar{k}}(u,t=0)}{u-ip/k}\, du}{D(k,ip)} \exp(pt)\, dp$$

for $\mathrm{Re}p \geq p_0$ (from the def of Laplace transform)

Now we find the poles of $D(k, ip) = 0$

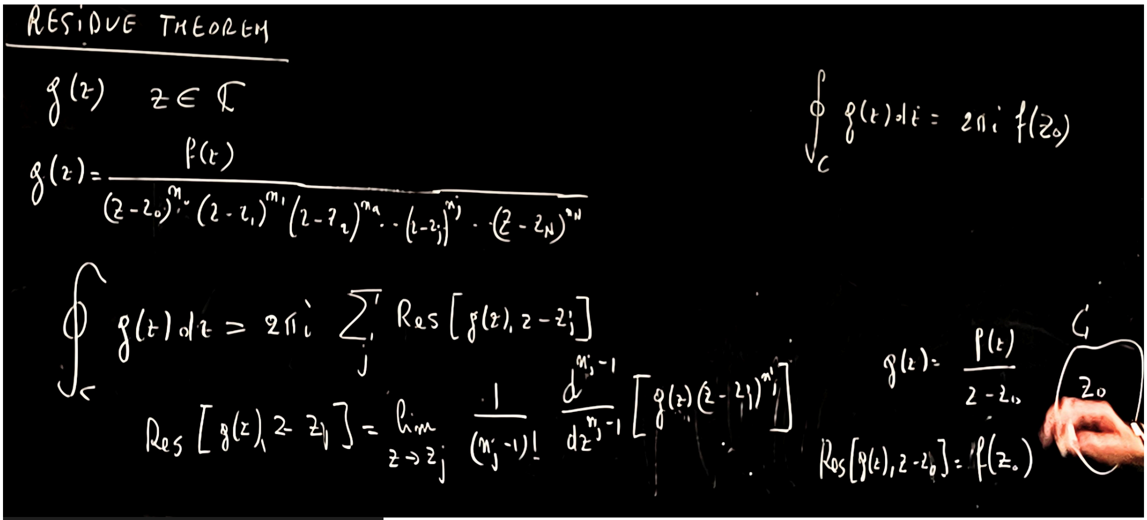
PIC - poles on the left of p0

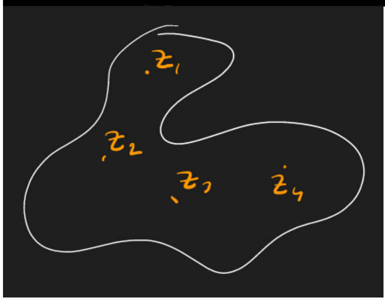


They are all on the left of p_0, because on the right of p_0 we have the validity of the Laplace Transform.

To use the residue theorem, we need to embrace al the poles, so we need to modify the domain of . We need to use the analytical continuation of the f.

PIC - residue theorem





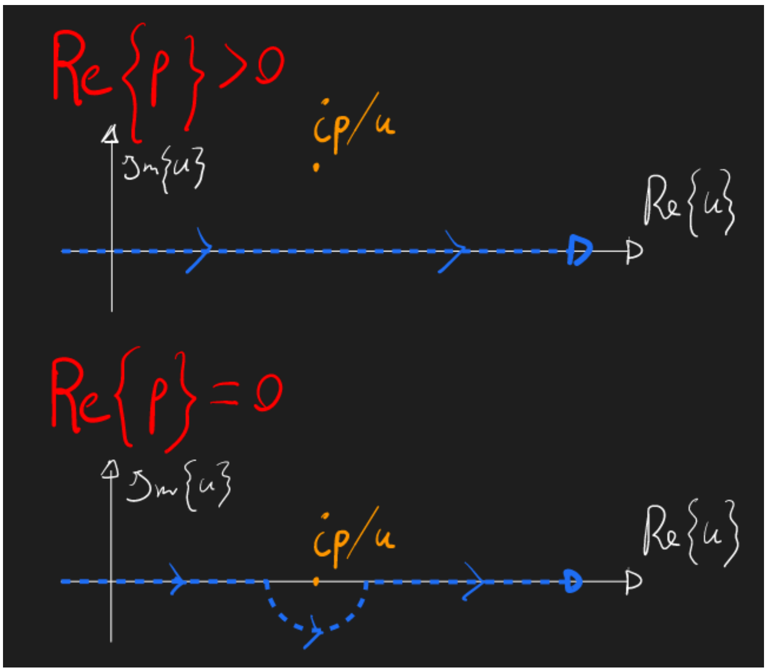
So we need the analytical continuation of $\hat{\phi}_{\bar{k}}(p)$, we call it

$$\hat{\Phi}_{\bar{k}}(p) = \begin{cases} \hat{\phi}_{\bar{k}}(p) & \text{Re}p \geq p_0 \\ \text{nalitical continuation of } \hat{\phi}_{\bar{k}}(p) & \text{Re}p < p_0 \end{cases}$$

We use the Landau prescription.
Both integrals can be written as

$$\int \frac{(u)}{u - ip/k} du, \qquad \text{with } (u) = \begin{cases} F_{\alpha 0}(u, t=0) & \text{for the numerator} \\ \partial F_{\alpha 0} / \partial u & \text{for the denominator} \end{cases}$$

PIC - how to avoid poles in integration



For the first case, $\text{Re}p \geq 0$ and we don't have any problems.

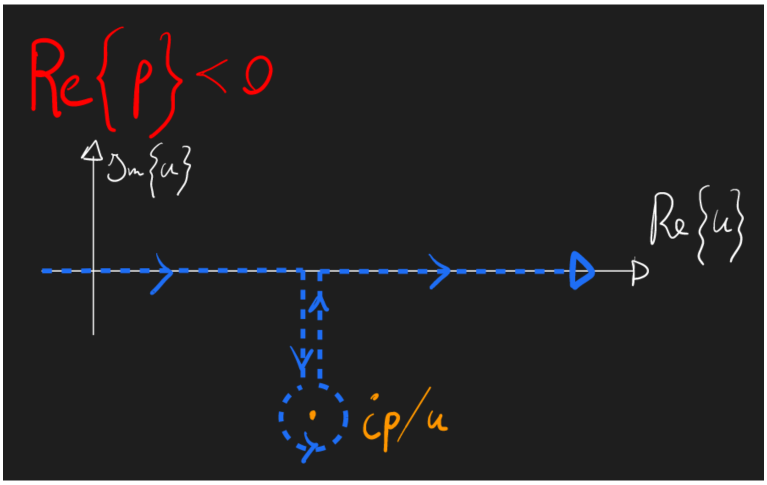
For the second case, $\mathrm{Re}p = 0$ and we need to modify the integration path.
We use

$$\int_{-\infty}^{\infty} \frac{(u)}{u - ip/k}\, du = \int_{-\infty}^{\infty} \frac{(u)}{u - ip/k}\, du + \pi i \cdot \left(\frac{ip}{k}\right)$$

and

$$\int_{-\infty}^{\infty} \frac{(u)}{u - ip/k}\, du = \lim_{\to 0} \left[\int_{-\infty}^{ip/k-} + \int_{ip/k+}^{\infty}\right] \frac{(u)}{u - ip/k}\, du$$

For the third case, $\mathrm{Re}p < 0$
PIC - 3rd case



In order to keep Φ analytic, we also have to move the integration path.
We use

$$\int_{-\infty}^{\infty} \frac{(u)}{u - ip/k}\, du = \int_{-\infty}^{\infty} \frac{(u)}{u - ip/k}\, du + 2\pi i \cdot \left(\frac{ip}{k}\right)$$

(a compact way to write this is $\int_{-\infty}^{\infty} \to \int$).

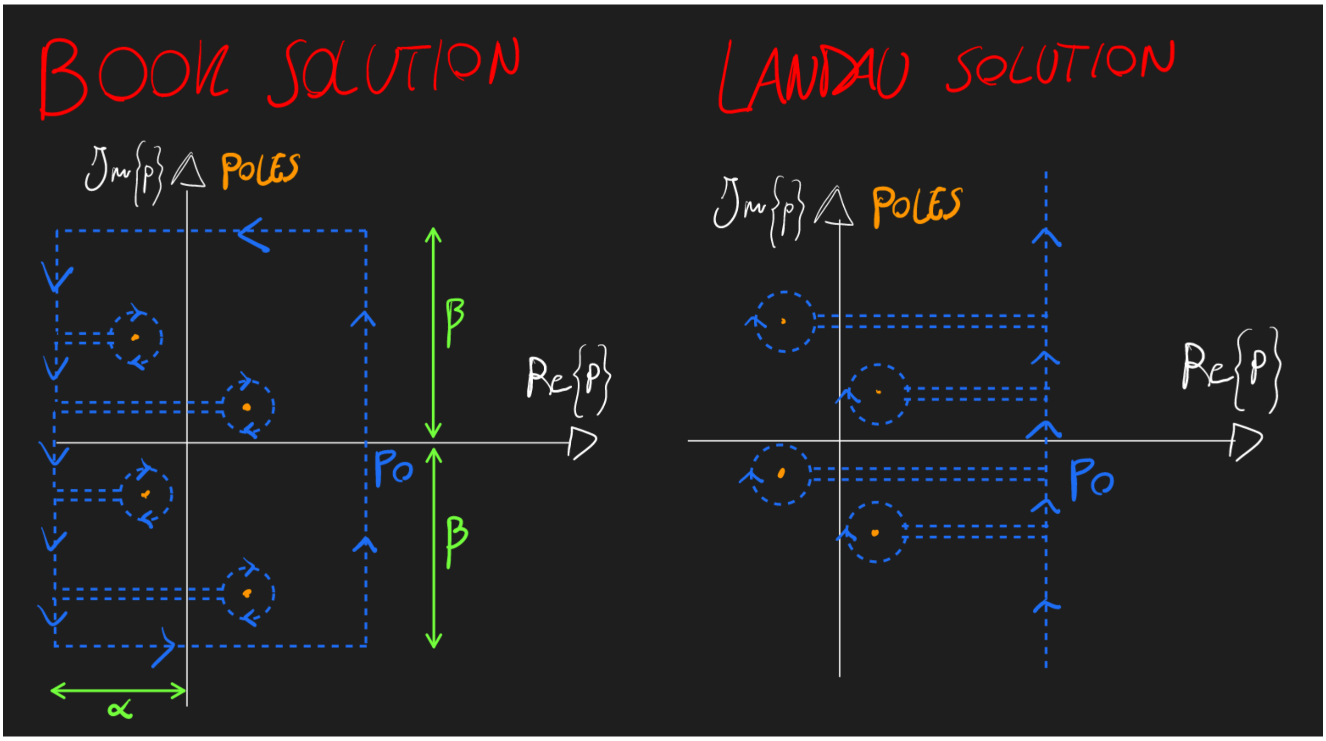
So we have

$$\phi_k(t) = \frac{1}{2\pi i} \int_{p_0 - i\infty}^{p_0 + i\infty} \frac{(-i)\frac{4\pi}{k} \sum_\alpha q_\alpha \int \frac{F_{\alpha \bar{k}}(u, t=0)}{u - ip/k}\, du}{D(k, ip)} \exp(pt)\, dp$$

for $\mathrm{Re}p \geq p_0$
and

$$D(k, ip) \equiv 1 - \sum_\alpha \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \int \frac{\partial F_{\alpha 0}(u)/\partial u}{u - ip/k}\, du$$

Now we follow the explanation by the book.
PIC - book approach vs landau approach



(Landau method breaks in the non-analiticy zone and the mathematical structure needs to be corrected)

In the book method we need

$$k^2\phi_{\bar{k}}(t) = \frac{1}{2\pi i}\left[\int_{p_0-i\infty}^{p_0-i} + \int_{p_0-i}^{-\alpha-i} + \int_{-\alpha-i}^{-\alpha+i} + \int_{-\alpha+i}^{p_0+i} + \int_{p_0+i}^{p_0+i\infty}\right](\cdots)\,dp + \text{poles}$$

Assumptions:

1. $\to \infty$
   then only $\int_{-\alpha-i}^{-\alpha+i}(\cdots)\,dp$ survives because for all the others $D(k, ip) \to \infty$ and they all tend to zero
2. long time limit (asymptotic)
   we neglect the short transient of perturbations and we focus on the long-term behaviour of the system.
   We know an upper limit for the remaining integral $\int_{-\alpha-i}^{-\alpha+i}(\cdots)\,dp \;\; \exp(-At^2)$ and we can also neglect it.

We are left with the contribution of the poles, which is

$$\sum_j {}_j \exp[p_j(k)t]$$

with

- $p_j(k)$ roots of $D(k, ip)$
- ${}_j = \lim_{p \to p_j}(p - p_j)\,\tilde{\Phi}_{\bar{k}}(p)$
  and

$$k^2\phi_{\bar{k}}(t) = \frac{1}{2\pi i}\left\{\cancel{\int_{-\alpha-i}^{-\alpha+i}(\cdots)\,dp} \; + 2\pi i\sum_j {}_j \exp[p_j(k)t]\right\} \approx \sum_j {}_j \exp[p_j(k)t]$$

Now using

$$ip_j(k) = \omega_j(k) \qquad \Longrightarrow \qquad p_j(k) = -i\omega_j(k)$$

we derive

$$D(k, \omega) = 0 \qquad \Longrightarrow \qquad \omega = \omega(k), \quad \omega \in \mathbb{C}$$

So

$$\phi_k(t) \ \sum_j {}_j \exp[-i\omega_j(k)t] =$$

$$= \sum_j {}_j \exp[-i(\omega_j(k) + i\omega_{Ij}(k))t] =$$

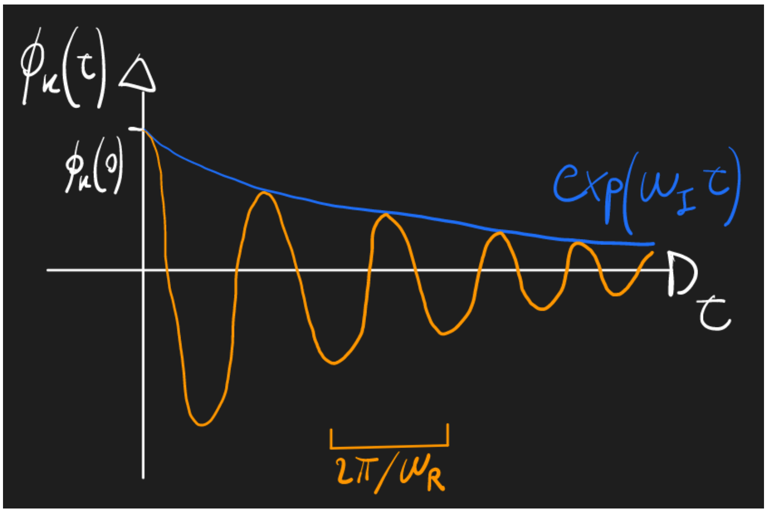
$$= \sum_j {}_j \exp[-i\omega_j(k)t] \exp[\omega_{Ij}(k))t]$$

where the last exponential (with the ω_{Ij} term) is responsible for the amplification or the damping of the wave. This factor is not predicted at all in fluid theory.

For $f_{\alpha 0}$ as a Maxwellian distribution at equilibrium

$$\omega_{Ij} < 0 \implies \text{Landau dampin}$$

PIC - landau damping



We have the energy paradox.
The system is isotropic so the energy is conserved and we could invert time.
Landau damping disperse energy, where does the energy go? If we reverse time, where would the energy come from?

Could, Malburg and O'Neil solved this paradox and showed where the energy goes using the **Plasma Echo Theory**.

We solve for ω

$$D(k, \omega) = 1 - \sum_\alpha \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \int \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega/k} du = 0$$

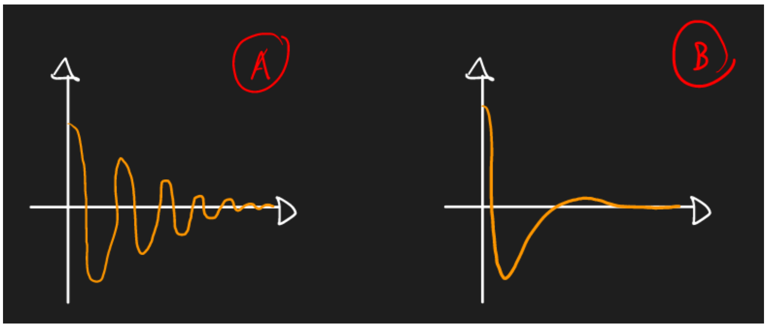
We will see that

$$D(k, \omega + i\omega_I) \approx D(k, \omega) + i\omega_I \frac{\partial D(k, \omega)}{\partial \omega}$$

in fact , we assume

$$|\omega_I / \omega| \ll 1$$

We see $|\omega_I/\omega| \ll 1$ (in picture A) compared to $|\omega_I/\omega| \;\; 1$ (in picture B)



We are interested in $\omega_I < 0$ because we cannot even observe a wave otherwise.
For $\omega_I > 0$ the linear approximation is not valid anymore and the wave grows in time exponentially.

We Taylor expand this in $\omega_I = 0$ and, using the Landau prescription (the poles are on the real axis)

$$D(k, \omega) = 1 - \sum_\alpha \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \int \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega/k} \, du \approx$$

$$\approx \underbrace{1 - \sum_\alpha \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega/k} \, du}_{D(k,\omega)} \; \underbrace{- \sum_\alpha \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot i\pi \frac{\partial F_{\alpha 0}(u)}{\partial u}\bigg|_{\omega/k}}_{D_I(k,\omega)}$$

and

$$D(k, \omega + i\omega_I) \approx \left[D(k, \omega) - \omega_I \frac{\partial D_I(k, \omega)}{\partial \omega} \right] + i \left[D_I(k, \omega) + \omega_I \frac{\partial D(k, \omega)}{\partial \omega} \right]$$

To find the zeros we need

$$\begin{cases} D(k, \omega) - \omega_I \dfrac{\partial D_I(k, \omega)}{\partial \omega} = 0 \\[2ex] D_I(k, \omega) + \omega_I \dfrac{\partial D(k, \omega)}{\partial \omega} = 0 \end{cases}$$

So in the first equation we neglect the second term since

$$\omega_I \frac{\partial D_I(k, \omega)}{\partial \omega} \quad \frac{\omega_I}{\omega} \implies D(k, \omega) = 0$$

from wich we can obtain the dispersion relation

$$\omega = \omega(k)$$

In the second equation, we keep the second term and we get

$$\omega_I = -\frac{D_I}{\partial D / \partial \omega}$$

otherwise we lose the Landau Damping effect.

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